Machine learning for coherent diffractive imaging with single-shot x-ray free electron laser pulses

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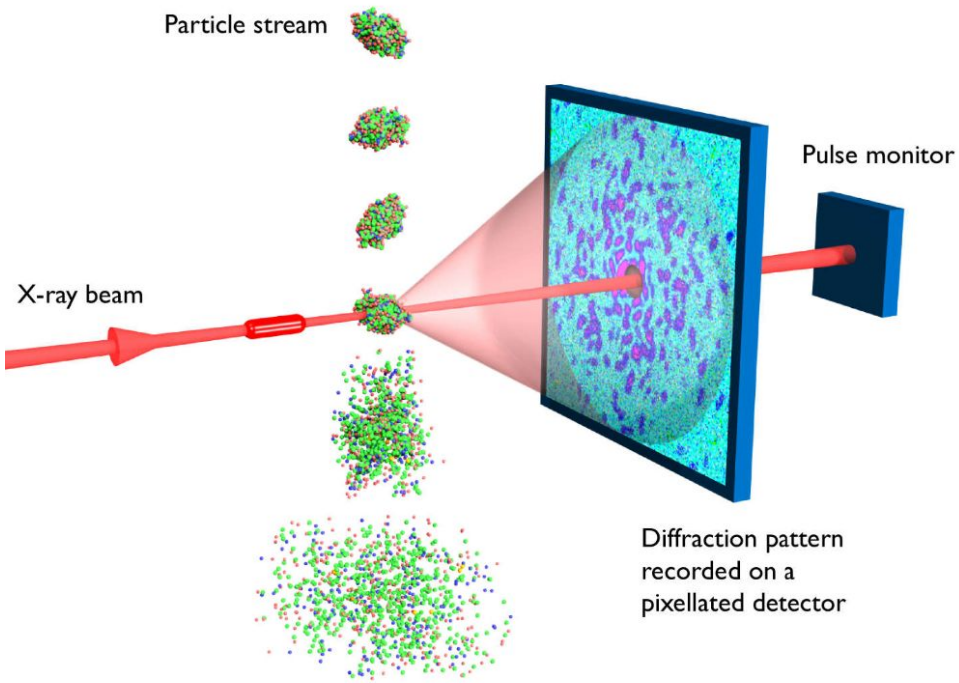
Coherent Diffractive Imaging

Lensless imaging is used to image particles less than 0.1 um in size, and the structure is reconstructed using computational methods.

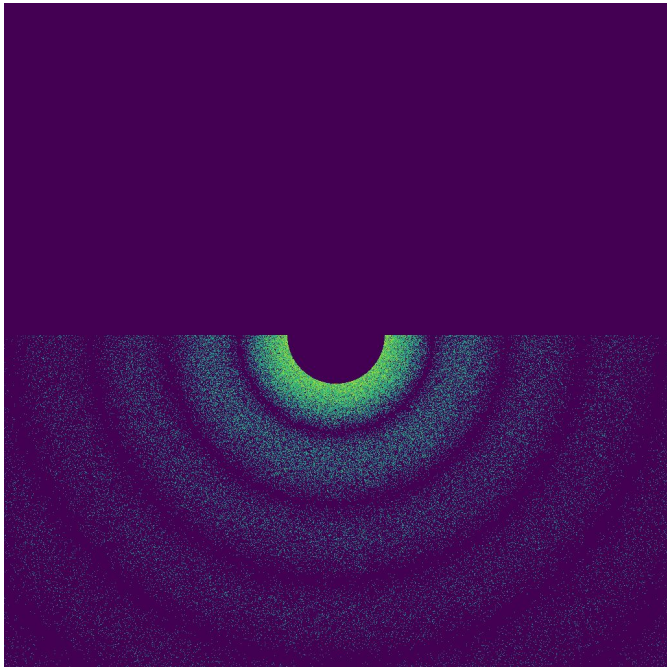
The phase information is lost during this process, and needs to be retrieved to reconstruct the particle.



Estimating size of a Spherical Particle: Determining Center

A major challenge in predicting the size of a sphere is determining the location of the center of the fringes.

Centers are determined by first estimating using a neural network, and then using Canny Edge detection to determine the precise location of the centers by projecting central lines between neighbouring points. The proximity to the neural network prediction is used as a heuristic for approximate location.



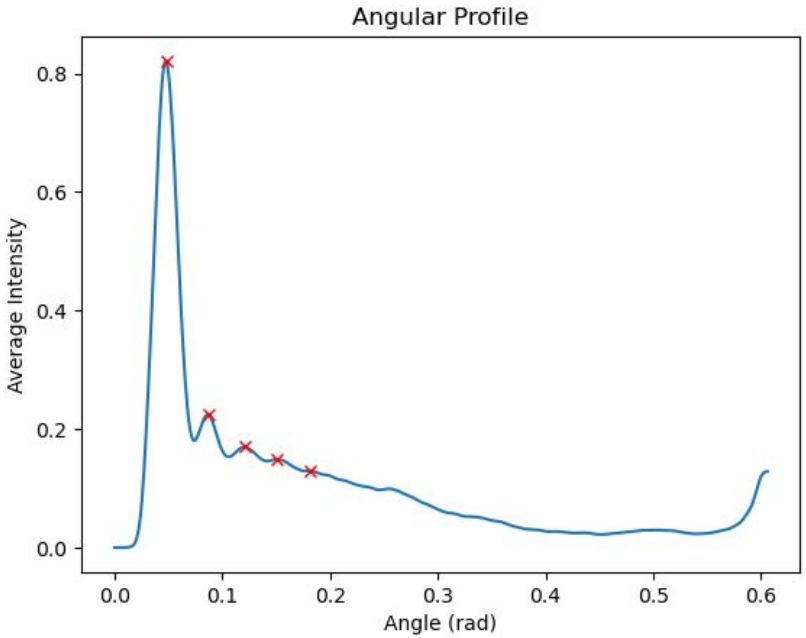
Data for simulated sphere with center of fringes at (0,0).

Estimating size of a Spherical Particle: Predicting Radius

First 5 nearest to center peaks were chosen as input to a neural network, that would predict the radius of the particle.

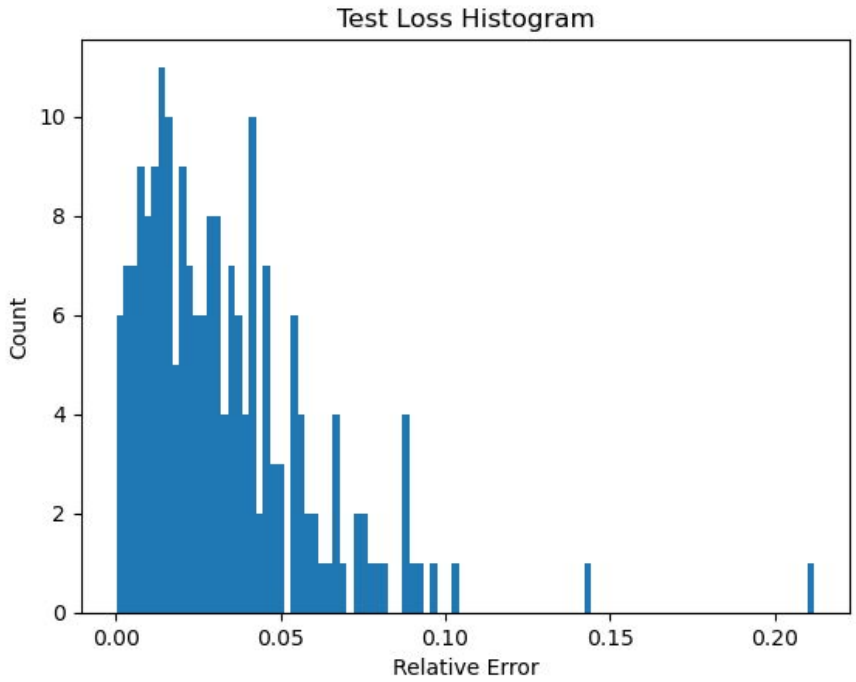
On testing, a 4% relative error was observed.

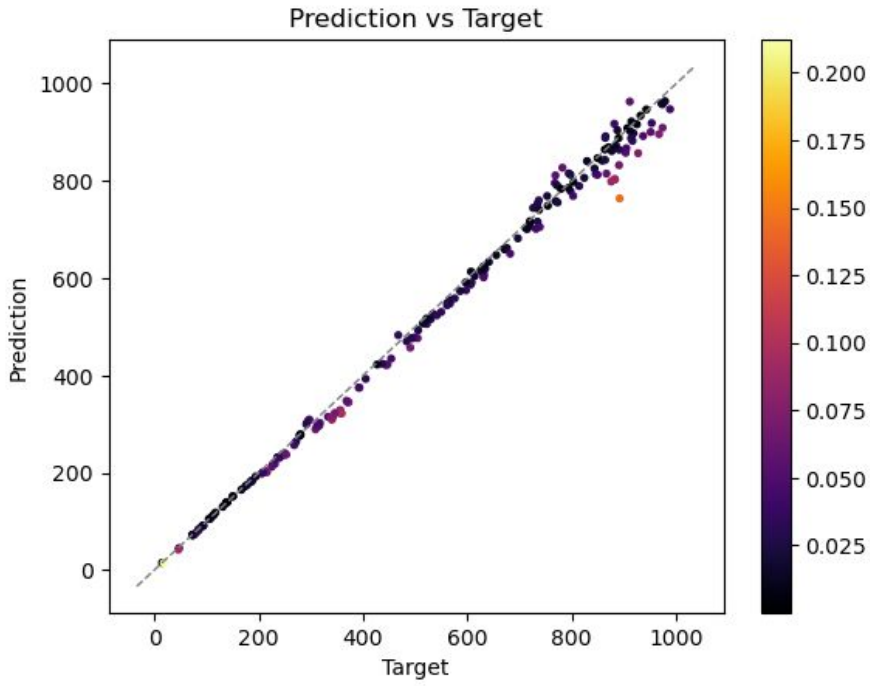
The data manifold repeats considerably, a “mixture of experts” model would work well to extend the domain of sizes of the particles that are predicted well.



Angular Profile for a sample Data point along with identified local maximas

Estimating size of a Spherical Particle: Model Performance



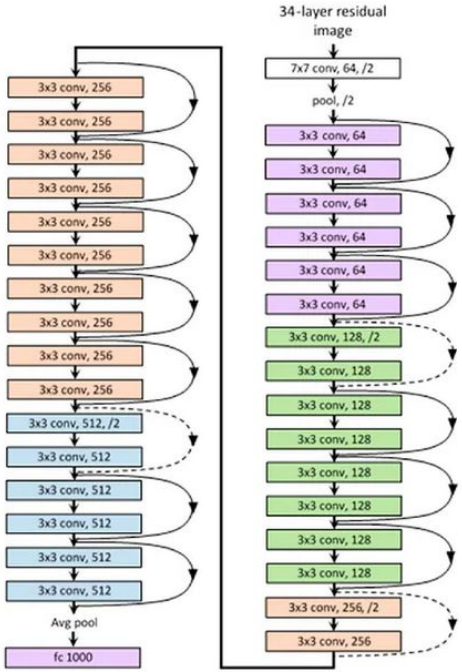


Estimating size of a Spherical Particle: End to End model

A few end to end neural networks were tested, that would take the diffracted image as the input, and predict the radius as output.

Even though the model performed slightly worse (6% error), the major issue was with the lack of interpretability of the model, and use of its results for reconstruction.

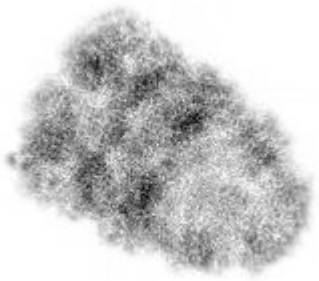
The training with larger models does show some double descent characteristics, but this can be only tested with exponentially larger models.



Binary Classification of a Particle

Along with determining the size of the particle, we tested performance of a model that would classify a particle as either GROEL-GROES or Poliovirus.

Encoding based similarity matching was also used with a CBR system to lookup the most similar known pattern and provided a single forward testing pass with orientation information, which did not perform significantly better than a simple neural network, but would perform better for larger number of samples.



GROEL-GROES

Poliovirus

Future Work

Test performance of other density distributions other than ellipsoids.

Design and test models with complex shapes, and reconstruction of complex shapes using spheres and other density distributions.

Assess the effectiveness of forward passing through simulation to steer predictions.

Acknowledgements

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Srijan S Bhat: Who is working alongside on the same project parallelly, and has been of great help in discussing, planning and implementation.

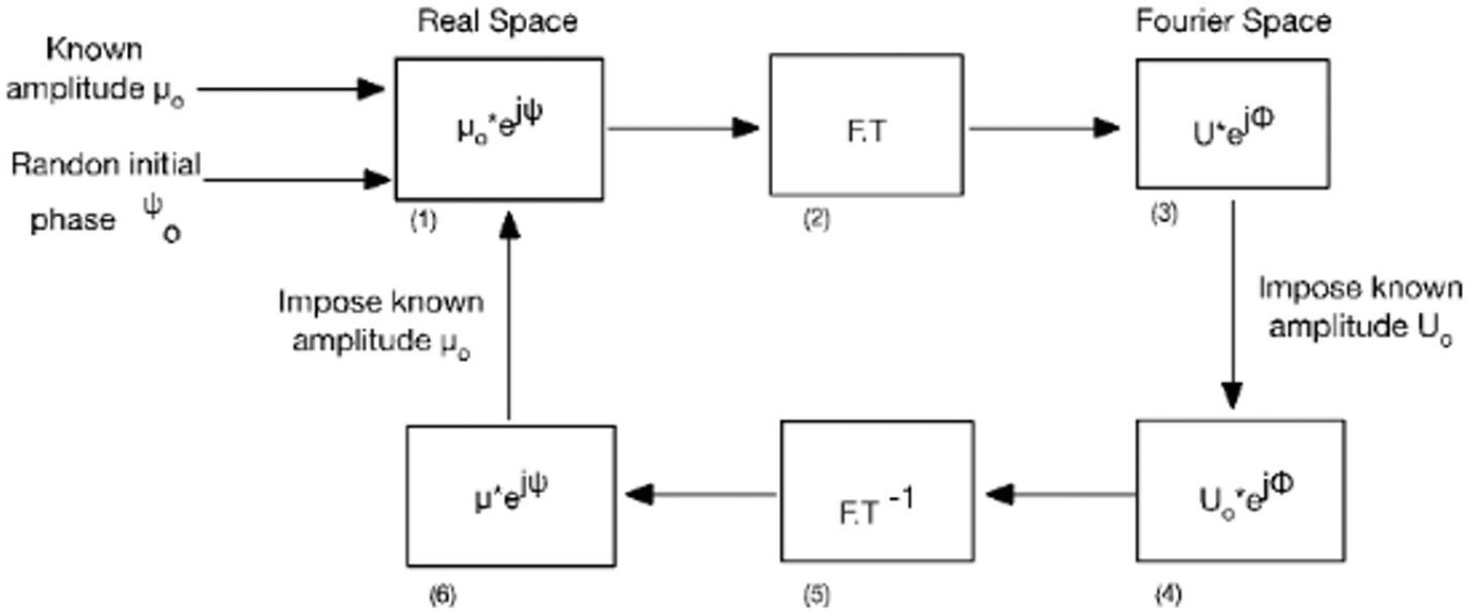
Thank You!

Things to do

- Experimental setup
- Existing methods
- Things that were tried
- Structure and size
- References
- Acknowledgements (ask regarding Prof. M Mudrich to Prof. Sivarama Krishnan)

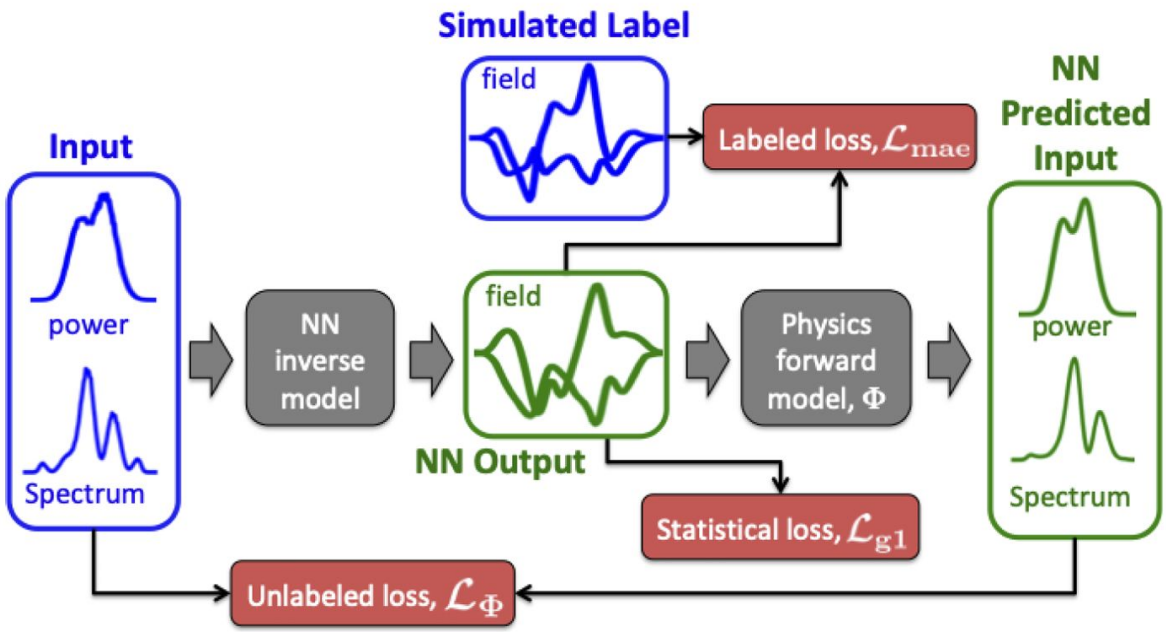
Supplementary Slides

Literature Review: Iterative Phase Retrieval



Shui K. An Iterative Approach to Reconstruct the Phase of a Field from an X-Ray Free Electron Laser Intensity Measurement. International Journal of High School Research. 2023 Oct 1;5(5).

Literature Review: ML upon Iterative Phase Retrieval



Ratner D, Christie F, Cryan JP, Edelen A, Lutman A, Zhang X. Recovering the phase and amplitude of X-ray FEL pulses using neural networks and differentiable models. Opt Express. 2021 Jun 21;29(13):20336-20352. doi: 10.1364/OE.432488. PMID: 34266125.